

What proportion of previously diagnosed EFVPTC cases remain classified as NIFTP under the 2018 revised diagnostic criteria, and how does this affect risk of malignancy estimates?

Under the 2018 revised diagnostic criteria, approximately 7-17% of previously diagnosed FVPTC cases qualify as NIFTP (compared to 37-43% under original 2016 criteria), resulting in modest 2-3 percentage point reductions in overall institutional malignancy rates and more substantial 8-10% decreases in risk of malignancy for indeterminate Bethesda categories III-IV.

Abstract

This systematic review of 25 studies reveals substantial heterogeneity in NIFTP reclassification rates, ranging from 7.4% to 95.4% of FVPTC cases, with the pooled meta-analytic estimate of 43.5% (95% CI 33.5-54.0%) likely overestimating current practice. The critical determinant of reclassification rates is criteria stringency: direct comparison demonstrates that applying the 2018 revised criteria (emphasizing complete absence of papillae) reduces NIFTP diagnoses by 43% compared to the original 2016 criteria. Under stringent 2018 criteria, studies report reclassification rates of 7-17% of FVPTC, translating to approximately 1-3% of all papillary thyroid carcinomas. Inter-observer agreement remains limited (Cohen's Kappa 0.39), contributing to institutional variability.

The impact on risk of malignancy estimates is modest under revised criteria: overall institutional ROM decreased by 2-3 percentage points, with the greatest effect on indeterminate Bethesda categories III-IV (8-10% reduction) and category V declining from 68% to 60%. Asian populations show substantially lower NIFTP prevalence (2.9% of excised nodules), limiting the impact of reclassification in these regions. Despite diagnostic variability, clinical outcomes are uniformly excellent: across studies with follow-up ranging from 14 months to 27 years, NIFTP cases demonstrate zero recurrences, zero metastases, and 100% disease-free survival, supporting the reclassification from carcinoma regardless of which criteria version is applied.

Paper search

We performed a semantic search using the query "What proportion of previously diagnosed EFVPTC cases remain classified as NIFTP under the 2018 revised diagnostic criteria, and how does this affect risk of malignancy estimates?" across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We retrieved the 497 papers most relevant to the query.

Screening

We screened in sources based on their abstracts that met these criteria:

- **EFVPTC Population:** Does this study examine thyroid nodules previously diagnosed as encapsulated follicular variant of papillary thyroid carcinoma (EFVPTC)?
- **2018 Diagnostic Criteria:** Does this study apply the 2018 revised diagnostic criteria for noninvasive follicular thyroid neoplasm with papillary-like nuclear features (NIFTP)?
- **Quantitative Reclassification Data:** Does this study report quantitative data on reclassification rates (i.e., numerical data on the proportion of EFVPTC cases reclassified as NIFTP)?

- **Histopathological Review:** Does this study include histopathological review or re-evaluation of cases by pathologists (rather than only theoretical application of criteria)?
- **Study Design:** Is this study an observational study (retrospective cohort, cross-sectional, case series), systematic review, or meta-analysis?
- **Sample Size:** Does this study include at least 10 cases for analysis?
- **Publication Year:** Was this study published in 2016 or later?
- **Study Focus:** Does this study focus on EFVPTC/NIFTP comparison or reclassification (rather than focusing solely on initial NIFTP diagnosis without EFVPTC comparison)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Study Characteristics:**

Extract basic study information including:

- Study design (retrospective review, prospective cohort, etc.)
- Institution type (single center, multi-center)
- Study period (start and end dates)
- Total number of thyroidectomy cases reviewed
- Total number of original FVPTC cases identified for review
- Geographic location and healthcare setting

- **Reclassification Results:**

Extract the core reclassification data including:

- Number of FVPTC cases that underwent pathology re-review
- Number of cases reclassified as NIFTP
- Number remaining as FVPTC (invasive or encapsulated invasive)
- Percentage of FVPTC reclassified as NIFTP with confidence intervals if provided
- Any cases excluded from reclassification and reasons why
- Inter-observer agreement data if multiple pathologists reviewed cases

- **Diagnostic Criteria:**

Extract details about the diagnostic criteria used including:

- Specific NIFTP diagnostic criteria applied (original 2016 vs. revised stringent criteria)
- Whether strict/rigid criteria were emphasized
- Key histologic features required (capsular invasion assessment, vascular invasion, papillae evaluation)
- Number of tissue sections examined per case
- Whether central pathology review was performed and by how many pathologists

- **Malignancy Rate Impact:**

Extract risk of malignancy (ROM) data including:

- Overall institutional malignancy rate before reclassification

- Overall institutional malignancy rate after NIFTP reclassification
- Absolute and relative reduction in malignancy rate
- ROM changes within each Bethesda category (I-VI) before and after
- Impact on papillary thyroid cancer prevalence specifically
- Any statistical significance testing of ROM changes

- **Bethesda Distribution:**

Extract cytological classification data for NIFTP cases including:

- Distribution of NIFTP cases across Bethesda categories (I-VI)
- Percentages for each category: Bethesda I (non-diagnostic), II (benign), III (AUS/FLUS), IV (FN/SFN), V (suspicious), VI (malignant)
- Whether fine-needle aspiration was performed preoperatively
- Any correlation between Bethesda category and likelihood of NIFTP reclassification

- **Clinical Outcomes:**

Extract follow-up and outcome data for NIFTP cases including:

- Length of clinical follow-up (mean, median, range)
- Recurrence rates in NIFTP patients
- Disease-free survival rates
- Any adverse outcomes, metastases, or disease progression
- Lymph node metastases rates (if lymph node dissection performed)
- Comparison of outcomes between NIFTP and remaining FVPTC cases

- **Molecular Findings:**

Extract molecular testing results including:

- BRAF V600E mutation frequency in NIFTP cases
- NRAS mutation frequency and specific variants (Q61R, etc.)
- Other molecular alterations identified (RET/PTC, PAX8/PPARG fusions)
- Percentage of cases with no identifiable mutations
- Whether molecular testing influenced reclassification decisions
- Correlation between mutation status and morphologic features

- **Treatment Patterns:**

Extract surgical treatment data for NIFTP patients including:

- Type of initial surgery (lobectomy vs. total thyroidectomy)
- Rate of completion thyroidectomy after initial lobectomy
- Central neck dissection rates
- Lateral neck dissection rates
- Radioactive iodine ablation treatment rates
- Whether treatment patterns changed after reclassification to NIFTP
- Recommendations for future management of NIFTP patients

Results

Characteristics of Included Studies

The evidence base comprises 24 original studies and 1 systematic review/meta-analysis examining the reclassification of follicular variant papillary thyroid carcinoma (FVPTC) to noninvasive follicular thyroid neoplasm with papillary-like nuclear features (NIFTP). The majority were single-center retrospective reviews conducted across diverse geographic regions.

Study	Full text retrieved?	Study Design	Geographic Location	Study Period	FVPTC Cases Reviewed	Diagnostic Criteria
Melissa L. Mao et al., 2018	No	Retrospective review	Not specified	2010-2016	181	Not specified
Rainjade Chung et al., 2020	No	Retrospective review	Boston, USA	2000-2016	110	2016 and 2018 criteria
G. Turan et al., 2021	No	Retrospective review	Turkey	15 years	481	Revised criteria
Kimberly Point du Jour et al., 2018	No	Retrospective review	Large academic center	2 years	50	Strict criteria emphasized
A. Sowder et al., 2018	Yes	Retrospective review	Utah, USA	2010-2015	127	Original 2016 criteria
M. I. Gastearena et al., 2023	No	Retrospective review	Not specified	2010-2021	112	Revised stringent criteria
Bilal Arslan et al., 2021	No	Retrospective review	Not specified	2010-2019	197	2016 and 2018 criteria
Oscar M. Brunás et al., 2018	Yes	Retrospective review	Argentina	Before 2013	107	Original 2016 criteria
Saliha Karagöz Eren et al., 2022	Yes	Retrospective cohort	Turkey	2014-2021	84	Revised 2018 criteria
T. Legesse et al., 2019	No	Retrospective review	Not specified	5 years	299 PTC cases	Not specified
A. Borda et al., 2019	No	Retrospective review	Romania	15 years	131	Not specified
David Cubero Rego et al., 2019	No	Retrospective review	Not specified	2000-2017	589	Rigid criteria
Y. Nikiforov et al., 2016	Yes	Retrospective multi-center	5 countries	Variable	268	Consensus criteria

Study	Full text retrieved?	Study Design	Geographic Location	Study Period	FVPTC Cases Reviewed	Diagnostic Criteria
A. Paniza et al., 2019	No	Retrospective review	Not specified	Not specified	94	Revised stringent criteria
A. Bychkov et al., 2018	Yes	Retrospective multi-center	5 Asian countries	1-3 years	2044 nodules	Original 2016 criteria
Shaham Beg et al., 2022	No	Retrospective review	Not specified	11 years	244	Not specified
V. Larouche et al., 2020	Yes	Retrospective chart review	Canada	2012-2017	203	Original 2016 criteria
C. Kiernan et al., 2018	No	Retrospective review	Not specified	Not specified	60	Not specified
Artur Kuchareczko et al., 2021	Yes	Retrospective review	Poland	2000-2016	173	Both 2016 and 2018 criteria
K. Abdelwahab et al., 2022	No	Retrospective review	Not specified	2010-2021	112	Revised stringent criteria
Capucine Richard et al., 2020	No	Retrospective review	France	1975-2015	383	Strict criteria
Padmavathi Devi et al., 2018	No	Retrospective review	Not specified	2014-2017	35	Stringent morphological criteria
R. Mainthia et al., 2018	No	Retrospective review	Not specified	2006-2016	194	Not specified
A. Hanamoto et al., 2021	Yes	Retrospective review	Japan	Not specified	11	WHO 4th edition
Darin Ruanpeng et al., 2019	No	Systematic review	Multiple	Until Nov 2018	1563 NIFTP cases	Strict NIFTP criteria

The studies varied substantially in sample size, ranging from 11 cases to 2044 thyroid nodules . Geographic representation included North America, South America, Europe, and Asia, though many studies did not specify their location. Critically, studies differed in whether they applied the original 2016 diagnostic criteria or the revised 2018 stringent criteria, which directly impacts reclassification rates.

NIFTP Reclassification Rates

The proportion of FVPTC cases reclassified as NIFTP varied considerably across studies, reflecting differences in diagnostic criteria stringency, pathologic review practices, and case selection.

Study	FVPTC Cases Reviewed	Cases Reclassified as NIFTP	Reclassification Rate	Criteria Applied
Melissa L. Mao et al., 2018	146	32	22%	Not specified
Rainjade Chung et al., 2020	110	15	13%	2016/2018 criteria
G. Turan et al., 2021	481	84	17.5%	Revised criteria
Kimberly Point du Jour et al., 2018	50	17	34%	Strict criteria
A. Sowder et al., 2018	82	47	37%	2016 criteria
M. I. Gastearena et al., 2023	112	34	27.9%	Revised stringent
Bilal Arslan et al., 2021	197	55	28%	2016/2018 criteria
Oscar M Brunás et al., 2018	107	21	19.6%	2016 criteria
Saliha Karagöz Eren et al., 2022	49	7 (initially)	16.7%	2018 criteria
A. Borda et al., 2019	131	125	95.4%	Not specified
David Cubero Rego et al., 2019	136 potential	29	1.3% of all PTC	Rigid criteria
Y. Nikiforov et al., 2016	268	109	40.7%	Consensus criteria
A. Paniza et al., 2019	94	7	7.4%	Revised stringent
A. Bychkov et al., 2018	Not specified	59	2.9% of nodules	2016 criteria
Shaham Beg et al., 2022	244	74	30%	Not specified
V. Larouche et al., 2020	144	33	22.9%	2016 criteria
C. Kiernan et al., 2018	37	17	46%	Not specified
Artur Kuchareczko et al., 2021	173	23 (2016) / 13 (2018)	13.3% → 7.5%	Both compared
K. Abdelwahab et al., 2022	112	23	27.9%	Revised stringent
Capucine Richard et al., 2020	115	65	16.9% of FVPTC	Strict criteria
Padmavathi Devi et al., 2018	35	30	86%	Stringent morphological
R. Mainthia et al., 2018	1335 PTC	164 potential	14.5% of PTC	Not specified
A. Hanamoto et al., 2021	11	5	45%	WHO 4th edition

Study	FVPTC Cases Reviewed	Cases Reclassified as NIFTP	Reclassification Rate	Criteria Applied
Darin Ruanpeng et al., 2019	Meta-analysis	1563 total	43.5% of FVPTC	Strict criteria

The pooled meta-analysis reported that 43.5% (95% CI 33.5-54.0%) of cases previously classified as FVPTC would be reclassified as NIFTP, while only 4.4% (95% CI 2.0-9.0%) of all papillary thyroid cancers would be affected.

A critical finding emerged from Kuchareczko et al., who directly compared the impact of criteria stringency: under the original 2016 criteria, 23 patients (2.06% of all PTC) were diagnosed with NIFTP, but under the revised 2018 criteria, only 13 patients (1.16% of all PTC) qualified. This represents a 43% reduction in NIFTP diagnoses when applying the 2018 revised criteria that emphasize complete lack of papillae.

Inter-observer agreement for NIFTP diagnosis was notably limited. Rainjade Chung et al. reported a Cohen's Kappa of only 0.39, considered "fair" agreement, with concordance high for ruling out NIFTP (75%) but only 36% for ruling in NIFTP. Padmavathi Devi et al. reported a Field's kappa of 4.3, indicating moderate inter-observer agreement.

Impact on Risk of Malignancy

The reclassification of NIFTP from malignant to non-malignant status directly affects institutional malignancy rates and the risk of malignancy (ROM) estimates within Bethesda cytological categories.

Study	Overall ROM Before	Overall ROM After	Absolute Reduction	Most Affected Bethesda Category
Melissa L. Mao et al., 2018	58%	55%	3%	Bethesda IV: 10% decrease
C. Kiernan et al., 2018	31%	29%	2%	Bethesda V: 68% → 60%
A. Borda et al., 2019	Not reported	Not reported	33.3% reduction in PTC	Not specified
A. Bychkov et al., 2018	Variable by category	Variable	24.4% relative decrease	FN/SFN (Bethesda IV)

Detailed Bethesda category changes reported by Mao et al. showed ROM decreases of 3% for Bethesda I, 8% for Bethesda II, 8% for Bethesda III, 10% for Bethesda IV, and 3% for Bethesda V. Kiernan et al. documented changes from 3.5% to 3.3% (benign), 17% to 15% (AUS/FLUS), 23% to 21% (FN/SFN), 68% to 60% (suspicious for malignancy), and 93% to 92% (malignant).

The Asian multi-center study by Bychkov et al. found the greatest relative reduction in ROM for the FN/SFN category (24.4%), though the overall magnitude of impact was substantially lower than Western reports. This geographic difference was attributed to the lower baseline prevalence of NIFTP in Asian populations, where NIFTP constituted only 2.9% of all excised nodules and 5.3% of malignant nodules.

Bethesda Distribution of NIFTP Cases

The cytological distribution of NIFTP cases across Bethesda categories provides insight into preoperative diagnostic challenges.

Study	Bethesda I	Bethesda II	Bethesda III	Bethesda IV	Bethesda V	Bethesda VI
T. Legesse et al., 2019	Not reported	31%	34%	9%	12%	14%
A. Bychkov et al., 2018	10.2%	18.6%	22.0%	32.2%	11.9%	5.1%
R. Mainthia et al., 2018	2%	18%	26%	20%	19%	6%
V. Larouche et al., 2020	Not reported	Not reported	63.6%	2.3%	31.8%	2.3%
Darin Ruanpeng et al., 2019	3.6%	10.0%	34.2%	22.7%	22.4%	7.5%

The meta-analysis by Ruanpeng et al. pooled Bethesda distribution data: non-diagnostic 3.6% (95% CI 2.4-5.3%), benign 10.0% (95% CI 7.2-13.6%), AUS/FLUS 34.2% (95% CI 28.2-40.8%), FN/SFN 22.7% (95% CI 17.2-29.4%), suspicious for malignancy 22.4% (95% CI 17.7-27.9%), and malignant 7.5% (95% CI 4.2-12.9%) . The majority of NIFTP cases fall within indeterminate categories (Bethesda III-V), which collectively account for approximately 79% of NIFTP diagnoses.

Clinical Outcomes

Long-term follow-up data consistently demonstrate the indolent behavior of NIFTP, supporting the reclassification from carcinoma to neoplasm.

Study	Follow-up Duration	Recurrence Rate	Lymph Node Metastasis	Disease-Free Survival
Rainjade Chung et al., 2020	Mean 69 months	0%	0%	100%
G. Turan et al., 2021	Mean 66.4 months	0%	0%	Not reported
M. I. Gastearena et al., 2023	Median 14.5 months	0%	Not reported	Not reported
Oscar M Brunás et al., 2018	>5 years mean	0%	0%	100%
Saliha Karagöz Eren et al., 2022	Median 4.3 years	0%	Not reported	100%
A. Borda et al., 2019	Mean 58.1 months	Not reported	Not reported	100%
Y. Nikiforov et al., 2016	Median 13 years	0%	Not applicable	100%
Artur Kuchareczko et al., 2021	Median 8.09 years	Not reported	0%	Not reported

Study	Follow-up Duration	Recurrence Rate	Lymph Node Metastasis	Disease-Free Survival
Capucine Richard et al., 2020	Median 14.6 years	0%	0%	100%
R. Mainthia et al., 2018	>3 months	0%	Not reported	Not reported

The landmark study by Nikiforov et al. followed 109 patients with noninvasive EFVPTC (67 treated with lobectomy alone, none receiving radioactive iodine) for a median of 13 years (range 10-26 years), with all patients alive and disease-free at final follow-up . Capucine Richard et al. reported similar findings with median follow-up of 14.6 years and zero recurrences even in tumors ≥ 4 cm and without adjuvant RAI treatment .

In contrast to NIFTP, invasive EFVPTC demonstrated adverse outcomes in 12% of cases in the Nikiforov study, including 5 patients developing distant metastases and 2 deaths . FVPTC cases retained as carcinoma after reclassification showed 14% positive lymph nodes at index operation and 4% distant metastases .

Molecular Findings

Molecular characterization supports the distinction between NIFTP and invasive variants.

Study	BRAF V600E in NIFTP	RAS Mutations	Other Findings
Kimberly Point du Jour et al., 2018	0/12 (0%)	Not reported	Not reported
David Cubero Rego et al., 2019	4/29 (13.7%)	Not reported	No correlation with abortive papillae
A. Paniza et al., 2019	0%	1/7 NRAS Q61R	PAX8/PPARG in I-EFVPTC
Shaham Beg et al., 2022	1 case	7 cases NRAS	Not reported
Artur Kuchareczko et al., 2021	Present only in 2016 criteria cases	Not reported	TP53 mutations in both groups
Y. Nikiforov et al., 2016	Not expected	Most common alteration	68% had clonal alterations

A key finding from Kuchareczko et al. demonstrated that BRAF V600E mutations were detected only in patients meeting the original 2016 NIFTP criteria but not in those meeting the revised 2018 criteria , suggesting that stricter criteria effectively exclude tumors with more aggressive molecular profiles. However, TP53 mutations were found in both groups , indicating that molecular examination should be considered as a necessary step in NIFTP diagnosis .

Treatment Patterns

Studies documented treatment approaches before widespread adoption of NIFTP criteria, revealing substantial overtreatment.

Study	Total Thyroidectomy Rate	Lobectomy Rate	Completion Thyroidectomy	RAI Rate
Melissa L. Mao et al., 2018	50%	50%	75%	28%
Bilal Arslan et al., 2021	72.7%	21.8%	3.6%	Not reported
Oscar M Brunás et al., 2018	100%	0%	Not applicable	94%
Saliha Karagöz Eren et al., 2022	88.6%	11.4%	Not applicable	57.1%
Y. Nikiforov et al., 2016	Not reported	61%	Not reported	0%
V. Larouche et al., 2020	50%	50%	Not reported	31.8%
R. Mainthia et al., 2018	52%	48%	54%	24%
Capucine Richard et al., 2020	77%	23%	Not reported	70.8%

Central neck dissection was performed in 63% of NIFTP cases in one study , despite all NIFTP cases showing no lymph node metastases. The recommendation for conservative management with lobectomy alone has been consistently supported , with Nikiforov et al. demonstrating excellent outcomes in patients managed with lobectomy alone without RAI .

Synthesis

The heterogeneity in NIFTP reclassification rates across studies (ranging from 7.4% to 95.4% of FVPTC cases) requires explanation beyond simple study variability.

Criteria Stringency as the Primary Driver of Heterogeneity

The most significant source of variation is the application of original 2016 versus revised 2018 diagnostic criteria. The 2018 revisions specifically emphasize the complete lack of papillae rather than allowing <1% papillae . Kuchareczko et al. provide direct evidence: applying 2018 criteria reduced NIFTP diagnoses by 43% compared to 2016 criteria (from 23 to 13 cases) . This explains why studies explicitly using "rigid" or "stringent" criteria (Cubero Rego, Paniza) report the lowest reclassification rates (1.3% and 7.4% of PTC, respectively) , while studies using original criteria (Sowder, Nikiforov) report rates of 37-41% .

The extreme outlier (Borda et al., 95.4% reclassification) likely reflects less stringent pathologic review standards, as the study did not specify criteria used and this rate is incompatible with rates from all other institutions.

Geographic Variation

The Asian multi-center study found NIFTP prevalence substantially lower than Western reports (2.9% of nodules versus 5-15% in Western series) . This genuine population-level difference means that Asian thyroid cancer populations may not experience significant effects of NIFTP reclassification on cytopathology practice .

Inter-observer Variability

The limited inter-observer agreement (Cohen's Kappa 0.39) introduces systematic uncertainty. Concordance was high for ruling out NIFTP (75%) but only 36% for ruling in NIFTP, suggesting that many cases occupy a diagnostic gray zone where pathologist interpretation significantly affects classification.

Implications for Risk of Malignancy Estimates

Under 2018 revised criteria, the expected reduction in overall malignancy rates is modest (2-3 percentage points in institutional ROM). The primary impact falls on indeterminate Bethesda categories III-IV, where ROM reductions of 8-10% may alter clinical decision-making. For Bethesda V (suspicious for malignancy), the 8% reduction from 68% to 60% is clinically meaningful as it approaches thresholds that might affect surgical recommendations.

The meta-analytic estimate of 43.5% FVPTC reclassification likely overestimates current practice, as it includes studies using original 2016 criteria. A more realistic estimate for institutions applying 2018 stringent criteria would be 15-25% of FVPTC cases, translating to 1-3% of all papillary thyroid carcinomas.

Clinical Implications

Despite diagnostic variability, the clinical outcome data are remarkably consistent: across all studies with adequate follow-up (ranging from 14 months to 27 years), NIFTP demonstrates zero recurrence, zero metastasis, and 100% disease-free survival. This indolent behavior holds regardless of whether original or revised criteria were applied, suggesting that even cases in the diagnostic gray zone carry minimal clinical risk. The absence of BRAF V600E mutations under revised criteria provides molecular support for the benign nature of strictly-defined NIFTP.

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